

- We are halfway through the semester!
⇒ Appropriately this means will now enter stage 2 of the course.

What we have been analyzing

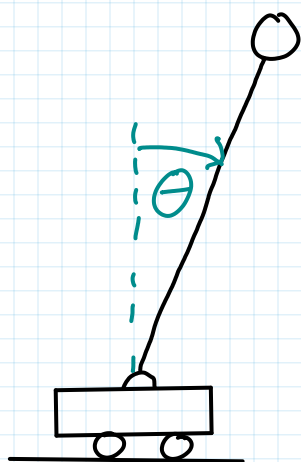
Control Systems

What we will look at now

- The halfpoint is a good opportunity for a "what happened so far"...

Recap:

Modelling



- How to describe a system with physics.
- Represent in this form:

$$\dot{x}(t) = f(x(t), u(t))$$

$$y(t) = g(x(t), u(t))$$

- We discussed the difference between
 - Inputs: $u(t)$
 - States: $x(t)$
 - Outputs: $y(t)$

Classification & Linearization

- We looked at these classifications:
 - Linear / Nonlinear
 - Causal / Non-causal
 - Time-variant / Time-invariant
 - Dynamic / Static
- In this course all systems are Linear & Time-invariant (LTI)
- We can say this, because systems can be linearized.

$$\begin{aligned}\dot{x}(t) &= f(x(t), u(t)) \\ y(t) &= g(x(t), u(t))\end{aligned}$$

↓ Jacobi
Matrices

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bx(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

⇒ Good approximation around
equilibrium point (x_e, u_e)

Time response & Stability

• Here we solved the diff' equation.

$$y(t) = Ce^{At}x_0 + C \int_0^t e^{A(t-\tau)} Bu(\tau) + Du(t)$$

Initial con-
dition response

Steady-state
response

- And saw, that a system is asymptotically stable if all:

$$\text{eigenvalues}(A) < 0.$$

Transfer Functions

- We can transform our systems into frequency-domain by using the Laplace-transform.

$$G(s) = C(sI - A)B + D$$

- Convolutions
 - Integrals
 - Derivatives
- } simple multiplications

- We looked at our new description of this realm:

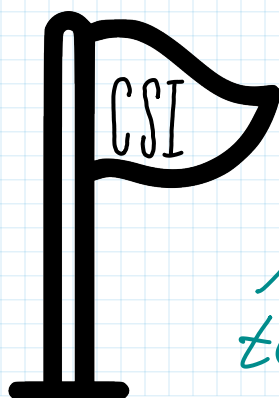
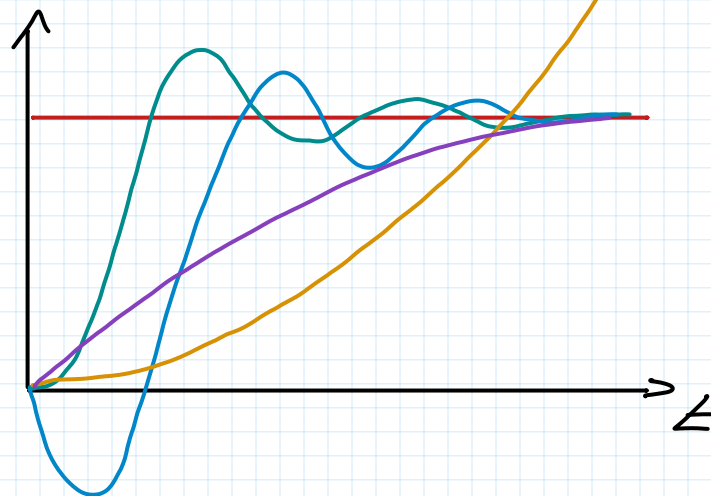
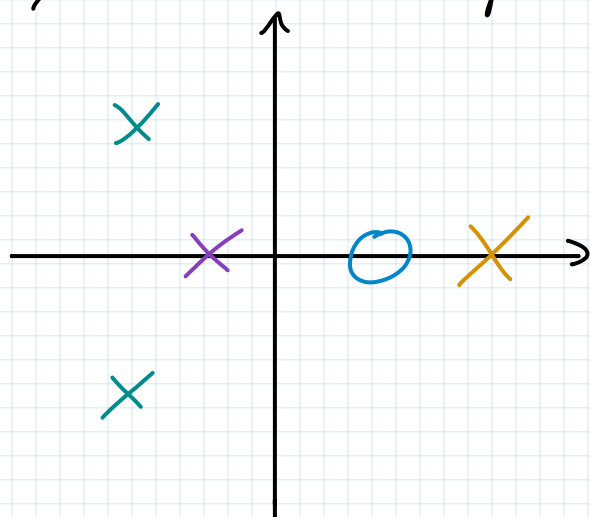
$$\Rightarrow \text{Complex-Frequency } s = \sigma + j\omega$$

Poles & Zeros

- We can characterize a system's behaviour by looking at its poles and zeros:

$$G(s) = k \frac{(s - z_1)(s - z_2) \dots (s - z_m)}{(s - p_1)(s - p_2) \dots (s - p_m)}$$

- The poles & zeros tell us how a system responds



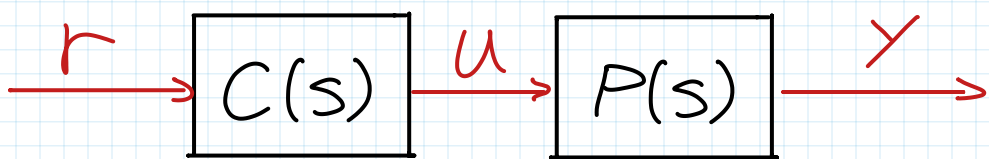
Checkpoint!

Now we go back,
to how it all began...

The initial problem

Model of system

r : reference
 u : input
 y : output



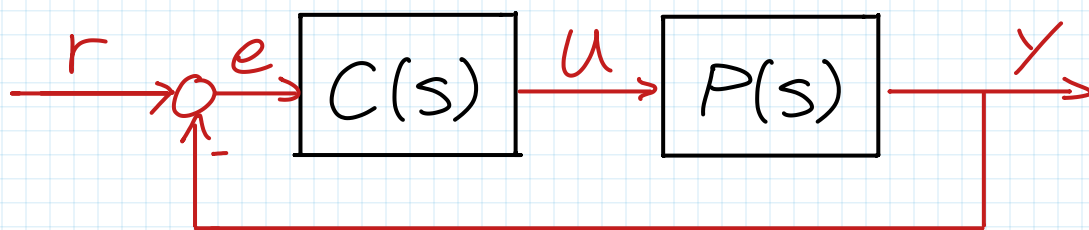
Our controller

- This we now call the open-loop $L(s)$

$$L(s) = C(s)P(s)$$

- Now, let's finally introduce feedback:

e : error



- The closed loop transfer function we derived like this:

$$y = C \cdot P \cdot e, \quad e = r - y$$

$$y = CP(r - y) \Rightarrow y(1 + CP) = CPr$$

$$y = \frac{CP}{1 + CP} r.$$

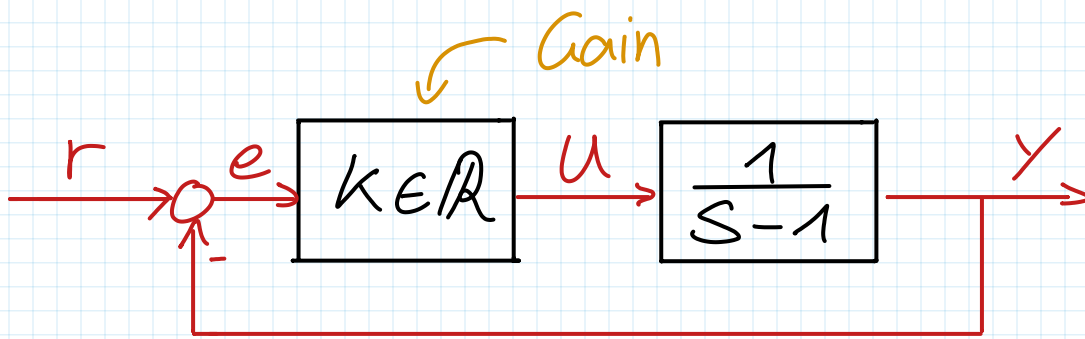
• We call this tf from $y \rightarrow r$ $T(s)$

$$T(s) = \frac{C(s)P(s)}{1 + C(s)P(s)} = \frac{L(s)}{1 + L(s)}$$

• $T(s)$ is also called complementary sensitivity function.

The essential question is now, how does feedback affect our system

Ex: Effect of proportional control

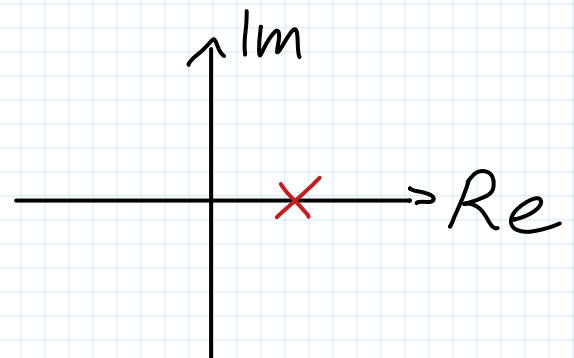


• Without feedback, the open-loop is:

$$L(s) = \frac{k}{s-1}$$

Pole-zero-Plot

$\Rightarrow$ Unstable, no matter our gain!!



• If we now close the loop, the tf becomes:

$$T(s) = \frac{L(s)}{1 + L(s)} = \frac{\frac{k}{s-1} \cdot (s-1)}{1 + \frac{k}{s-1} \cdot (s-1)}$$

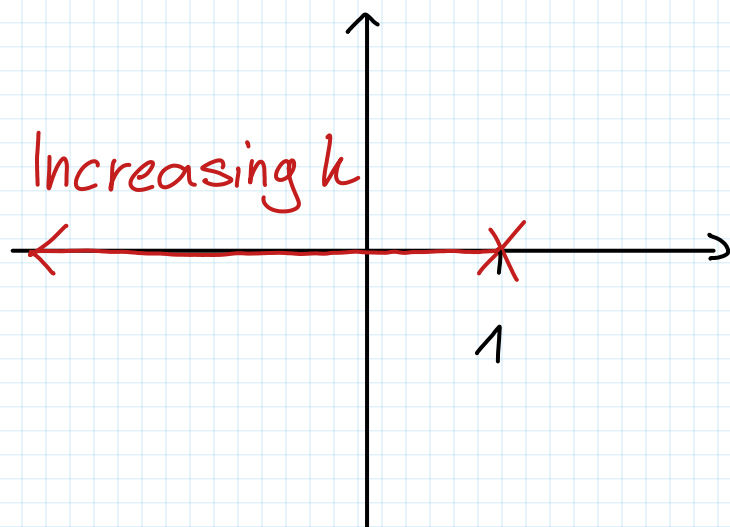
$$= \frac{k}{s+k-1}$$

Where are the poles now?

• denominator $\stackrel{!}{=} 0$

$$s_p + k - 1 = 0 \quad \Rightarrow \quad s_p = 1 - k$$

$\Rightarrow$ It depends on k !



In this case, the closed loop becomes stable for $k \in (1, \infty)$

This is our next important plot...

A plot that shows the position of poles and zeros over different gains k , is called a:

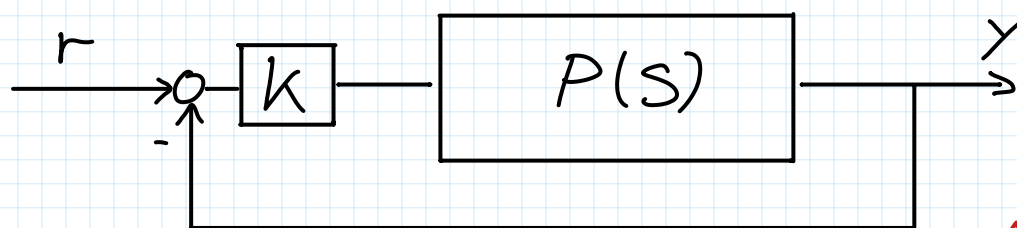
Root-Locus Plot

- To draw a root locus plot given a plant $P(s)$ you must follow a set of (pretty devious) rules...

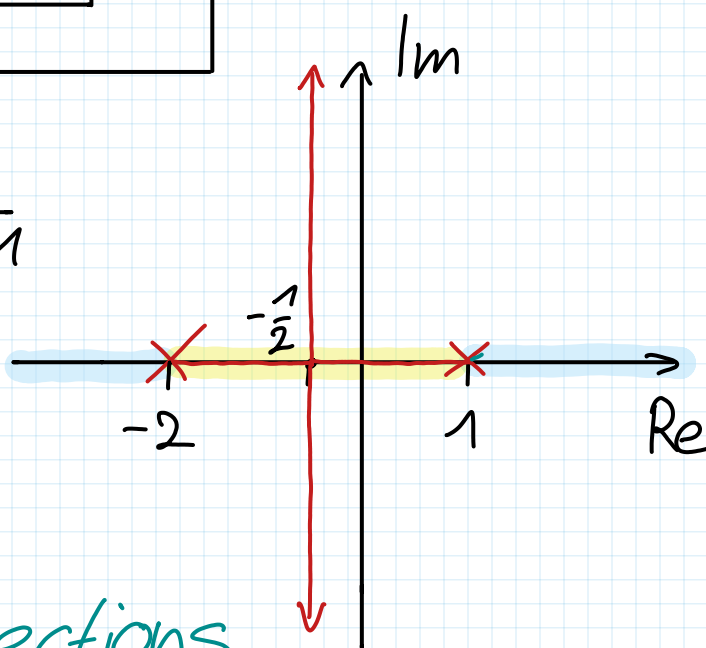
Rule Name	Description
Definitions	- The loop gain is $KG(s)H(s)$ which can be rewritten as $KN(s)/D(s)$. $N(s)$, the numerator, is an m^{th} order polynomial; $D(s)$ is n^{th} order. $N(s)$ has zeros at z_i ($i=1..m$); $D(s)$ has them at p_i ($i=1..n$). - The difference between n and m is q, so $q=n-m$.
Symmetry	The locus is symmetric about real axis (i.e., complex poles appear as conjugate pairs).
Number of Branches	There are n branches of the locus, one for each pole of the loop gain.
Starting and Ending Points	The locus starts (when $K=0$) at poles of the loop gain, and ends (when $K \rightarrow \infty$) at the zeros. Note: there are q zeros of the loop gain as $ s \rightarrow \infty$.
Locus on Real Axis	The locus exists on real axis to the left of an odd number of poles and zeros.
Asymptotes as $ s \rightarrow \infty$	If $q > 0$ there are asymptotes of the root locus that intersect the real axis at $\sigma = \frac{\sum_{i=1}^n p_i - \sum_{i=1}^m z_i}{q}$, and radiate out with angles $\theta = \pm r \frac{180}{q}$, where $r=1, 3, 5, \dots$
Break-Away and -In Points on Real Axis	There are break-away or -in points of the locus on the axis wherever $N(s)D'(s) - N'(s)D(s) = 0$.
Angle of Departure from Complex Pole	Angle of departure from pole p_j is $\theta_{\text{depart}, p_j} = 180^\circ + \sum_{i=1}^m \angle(p_j - z_i) - \sum_{i=1, i \neq j}^n \angle(p_j - p_i)$
Angle of Arrival at Complex Zero	Angle of arrival at zero z_j is $\theta_{\text{arrive}, z_j} = 180^\circ - \sum_{i=1, i \neq j}^m \angle(z_j - z_i) + \sum_{i=1}^n \angle(z_j - p_i)$
Locus Crosses Imaginary Axis	Use Routh-Horwitz to determine where the locus crosses the imaginary axis.
Determine Location of Poles, Given Gain "K"	Rewrite characteristic equation as $D(s) + KN(s) = 0$. Put value of K into equation, and find roots of characteristic equation. (This may require a computer)
Determine Value of "K", Given Pole Locations	Rewrite characteristic equation as $K = -\frac{D(s)}{N(s)}$, replace "s" by the desired pole location and solve for K . Note: if "s" is not exactly on locus, K may be complex, but the imaginary part should be small. Take the real part of K for your answer.

Let's ignore these for a second and establish the rules visually...

Some examples:



$$P(s) = \frac{1}{(s+2)(s-1)} = \frac{1}{s^2 + s - 1}$$



1) Draw open loop poles & zeros

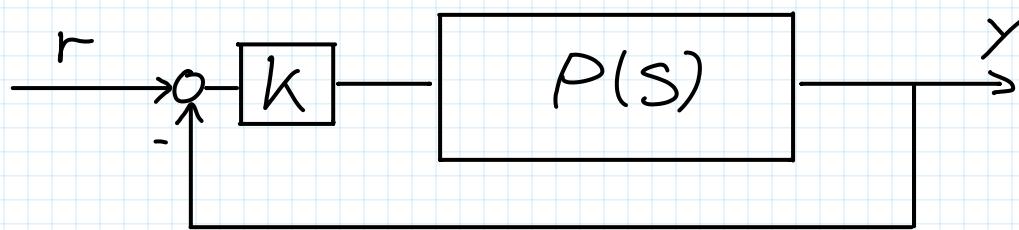
2) Identify filled/unfilled sections on the real axis

- ∞ to first pole/zero: No line, After that, alternate.

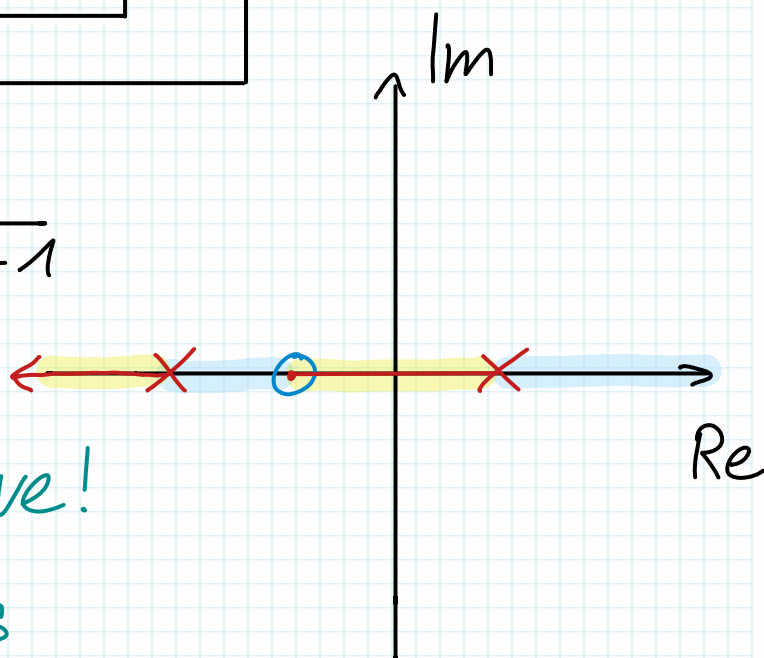
3) Poles meet in the "mass center"

4) Poles leave real axis at 90° angle

Ok, but what if we add zeros?



$$P(s) = \frac{s+1}{(s+2)(s-1)} = \frac{s+1}{s^2+s-1}$$

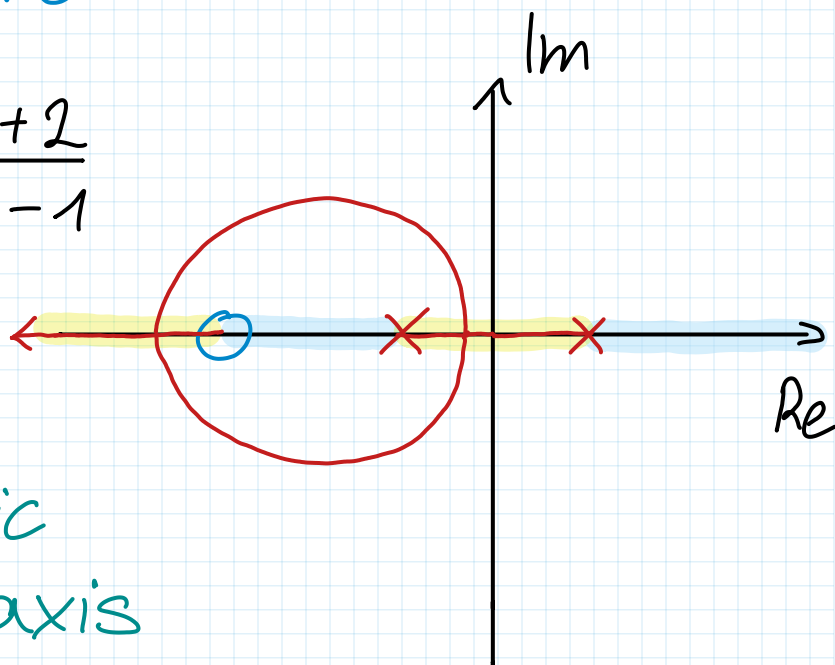


5) Zeros do not move!

6) Poles go to zeros

Watch what happens if we swap the left pole and zero:

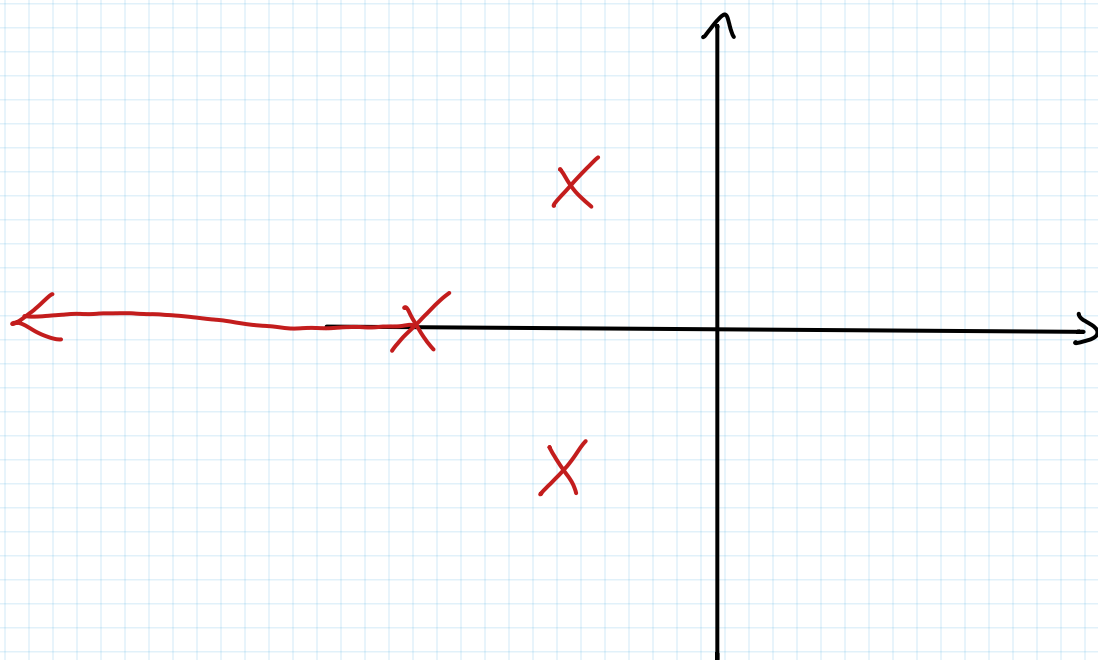
$$P(s) = \frac{s+2}{(s+1)(s-1)} = \frac{s+2}{s^2-1}$$



7) Locus is symmetric about the real axis

• The last rules are unfortunately harder...

$$P(s) = \frac{1}{(s+2)(s+1+j)(s+1-j)} = \frac{1}{s^3 + 4s^2 + 6s + 4}$$



8) Poles go along a asymptote, if there are more poles than zeros. We find the angles of the asymptote with the asymptote rule:

$$\theta_k = \frac{(2k+1) \cdot 180^\circ}{n-m}$$

$$k = 0, 1, 2, \dots, (n-m-1)$$

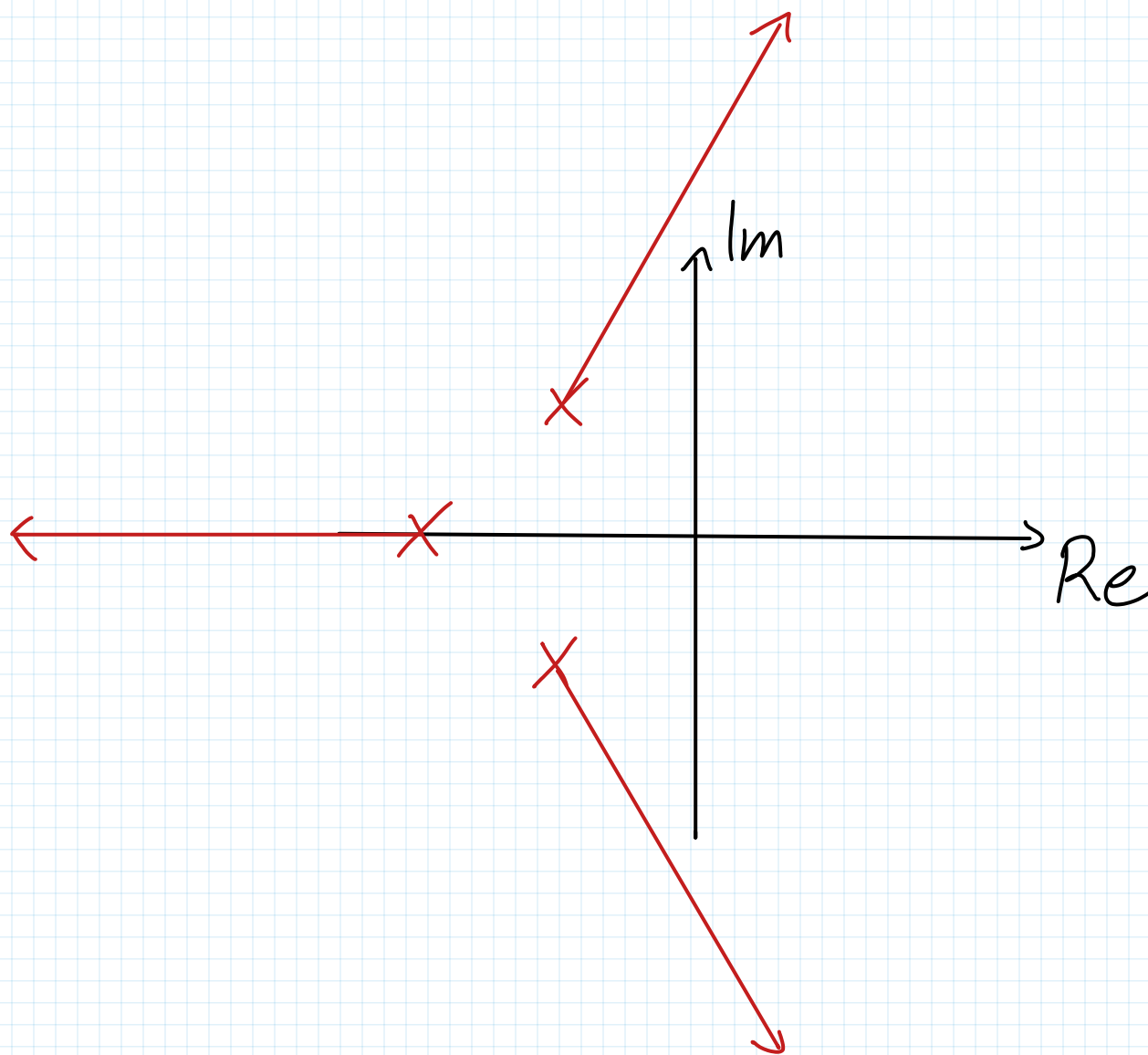
$$n \hat{=} \# \text{ poles}$$

$$m \hat{=} \# \text{ zeros}$$

Here:

$$\theta_1 = \frac{180^\circ}{3-0} = 60^\circ, \quad \theta_2 = \frac{3 \cdot 180^\circ}{3-0} = 180^\circ$$

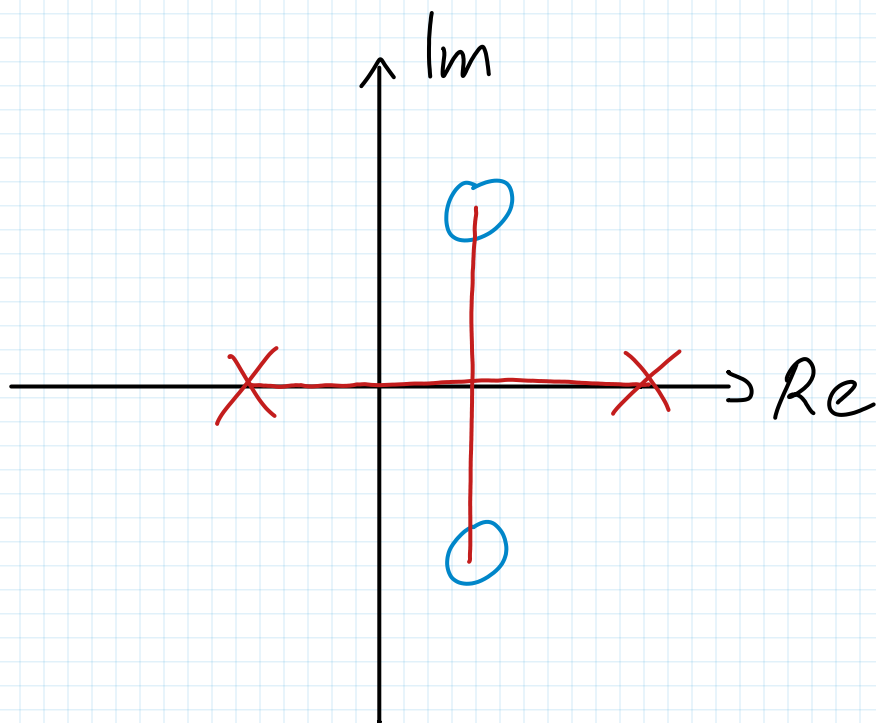
$$\theta_3 = \frac{5 \cdot 180^\circ}{3-0} = 300^\circ = -60^\circ$$



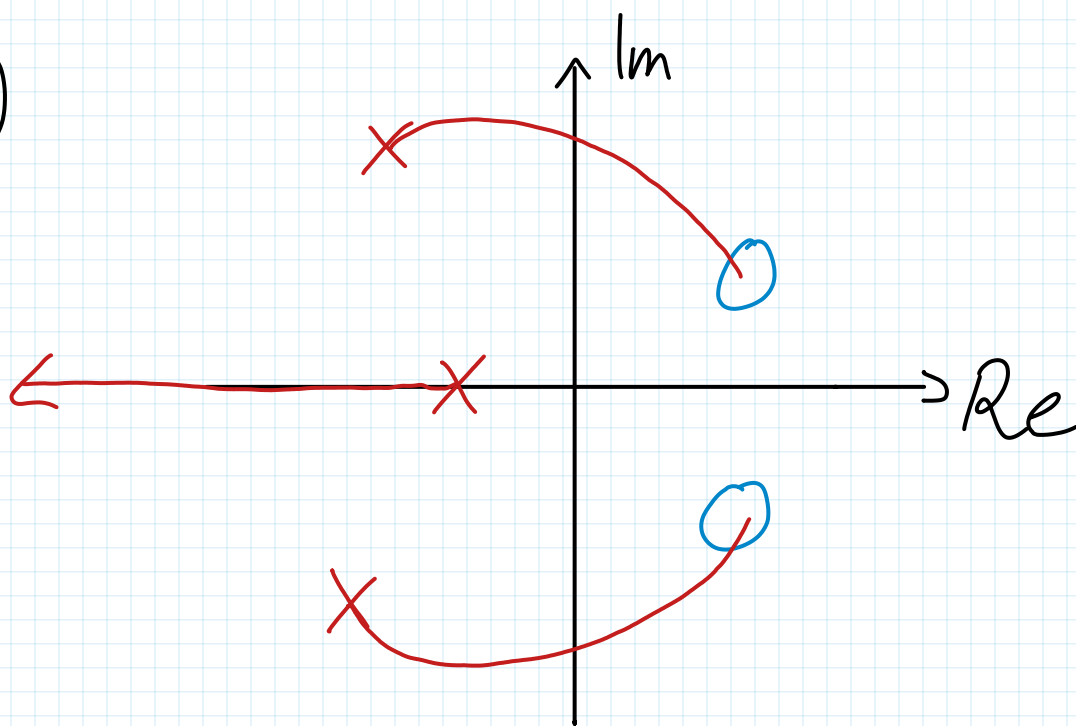
Intermediate quiz!!

Draw the root loci for these plots:

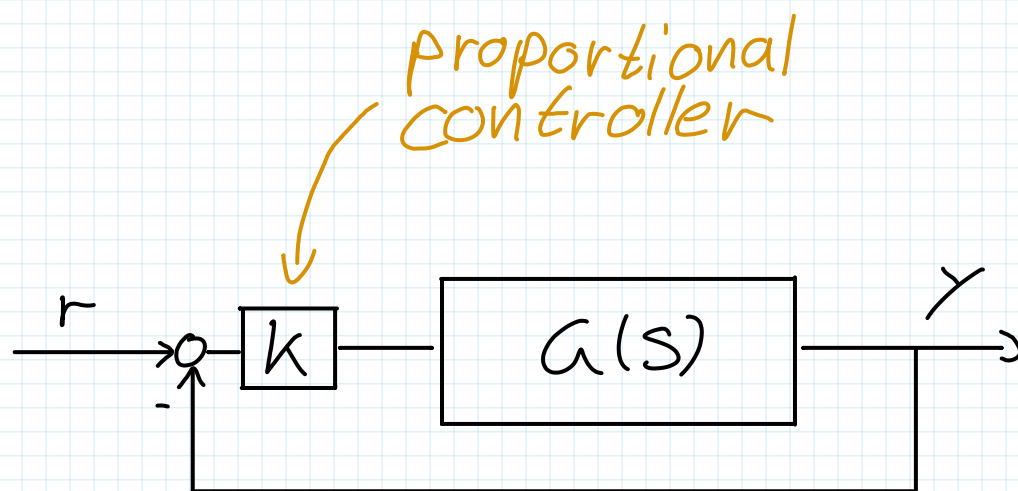
Q1)



Q2)



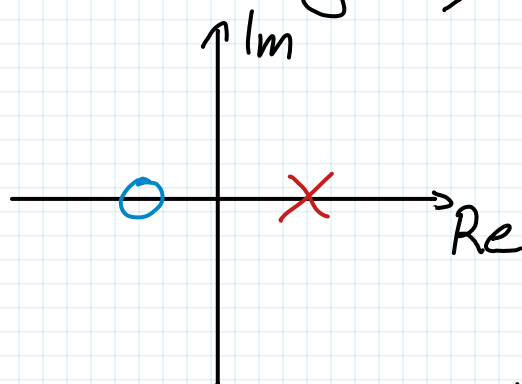
- So only by multiplying the signal that goes into the plant with an amplification (or gain) k , we have a lot of control over the systems behaviour!



How can we understand the root locus intuitively?

Ex: Consider the following system:

$$G(s) = \frac{(s+1)}{(s-1)}$$



- Now let's add a proportional controller k and look at the closed loop.

• TF of closed loop:

$$\begin{aligned}
 T(s) &= \frac{L(s)}{1 + L(s)} = \frac{k \cdot \frac{(s+1)}{(s-1)}}{1 + k \cdot \frac{(s+1)}{(s-1)}} \\
 &= \frac{k \cdot \frac{(s+1)}{(s-1)}}{\frac{(s-1) + k(s+1)}{(s-1)}} = \frac{k(s+1)}{(s-1) + k(s+1)}
 \end{aligned}$$

• Analyzing this shows us why the plotting rules work:

The zeros stay in the same spot $\forall k$

$$\frac{k(s+1)}{(s-1) + k(s+1)}$$

Increasing k means the open-loop zero becomes a more important closed-loop pole.

=> This is why the closed-loop poles move towards the open-loop zeros!

Now we know everything to tackle root-locus exam questions!

Problem: Consider the closed-loop system shown in Figure 11, where L represents a linear time-invariant system and $k \in \mathbb{R}$. The root locus of L for $k > 0$ is shown in Figure 12.

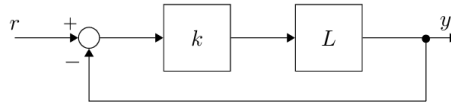


Figure 11: Standard feedback architecture. System L and proportional gain k .

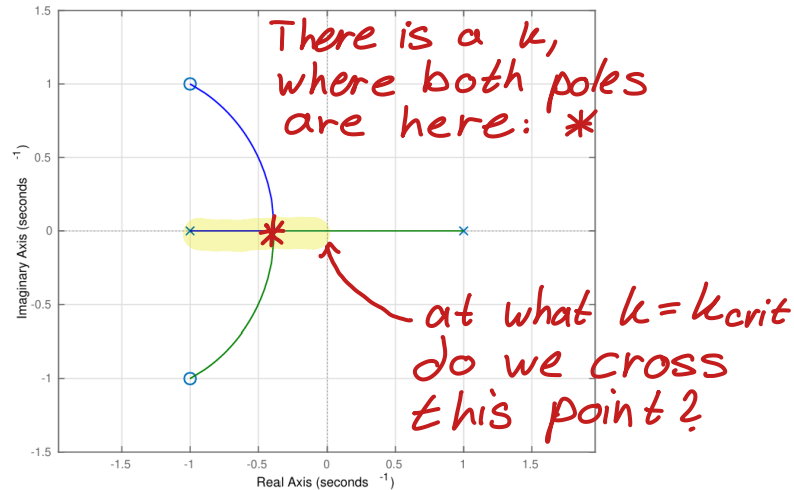


Figure 12: Root Locus of L . Poles are represented by $\times$ symbols, zeros by o symbols.

Q24 (0.5 Points) Mark the correct answer.
The open-loop system L is asymptotically stable.

☐ True ☒ False

Q25 (0.75 Points) Mark the correct answer.
There exist two gains, k_1 and k_2 , with $0 < k_1 < k_2$, such that for all $k \in [k_1, k_2]$, the closed-loop system is stable, and the closed-loop step response does not exhibit any overshoot.

☒ True ☐ False

Q24) Condition for asymptotic stability:

$\Rightarrow$ All poles < 0 .

Here: $p_1 = -1, p_2 = 1$

$\swarrow$ open-loop unstable!!

Q25) Condition for stability:

$\Rightarrow$ All poles ≤ 0

Condition for no overshoot:

$\Rightarrow$ All poles on real axis ($\in \mathbb{R}$)

So we must evaluate, if there is a range of k , where both poles are in this realm.

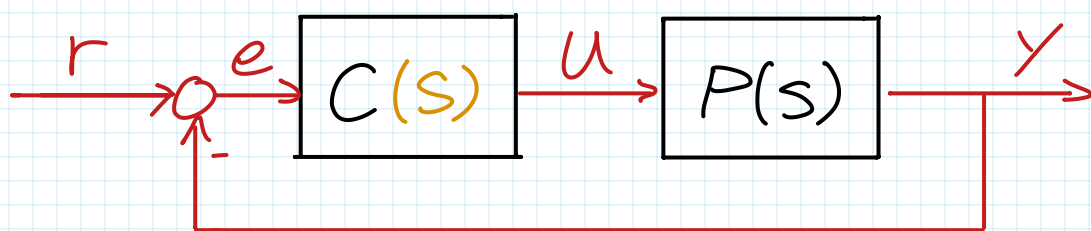
$\Rightarrow$ We must use rule #3:

Before branching off real axis, poles meet in the "mass center."

Now we understand the first (and most complex) of three main plots we will look at.

• We see now, how powerful a simple scalar as a controller can be.

But most of the time, our controller will be a **dynamic** system itself!



• A typical $C(s)$ design is combining these three components

- **P**roportional action: k_P
- **I**ntegral action: $k_I \cdot \frac{1}{s}$
- **D**erivative action: $k_D \cdot s$

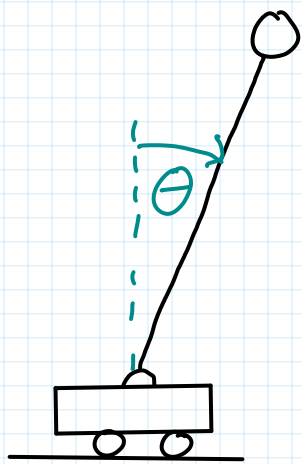
⇒ This is called a **PID-controller**

$$C_{PID}(s) = k_P + k_I \cdot \frac{1}{s} + k_D \cdot s$$

• Now we choose our gains k_P, k_I, k_D , and write the total system $L(s)$ as a function of just one gain k_{re} by bringing $L(s)$ into root-locus-form:

$$L(s) = \frac{k_{re} \cdot (s - z_1)(s - z_2) \dots (s - z_m)}{s^q (s - p_1)(s - p_2) \dots (s - p_{n-q})}$$

Ex: Inverted Pendulum



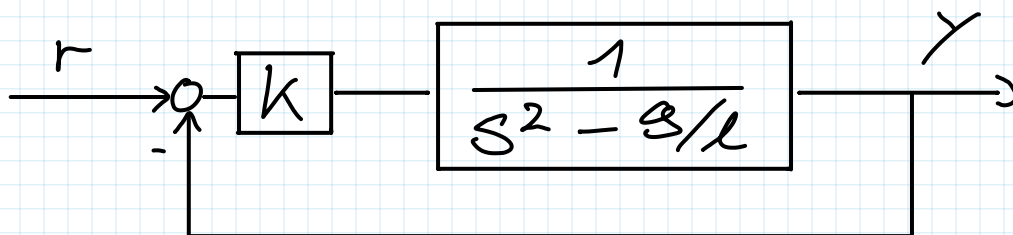
Linearized EoM:

$$m l^2 \ddot{\theta} = m g l + u$$

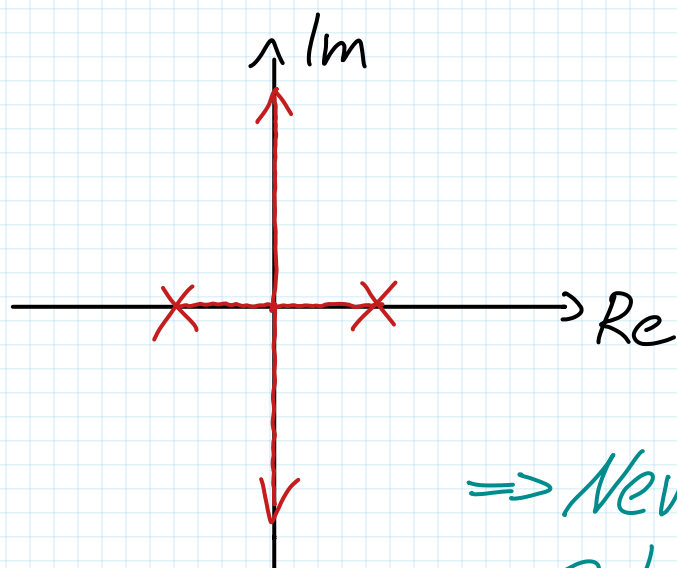
In s -domain this reads:

$$P(s) = \frac{1}{s^2 - g/l}$$

Let's try and add a p -controller



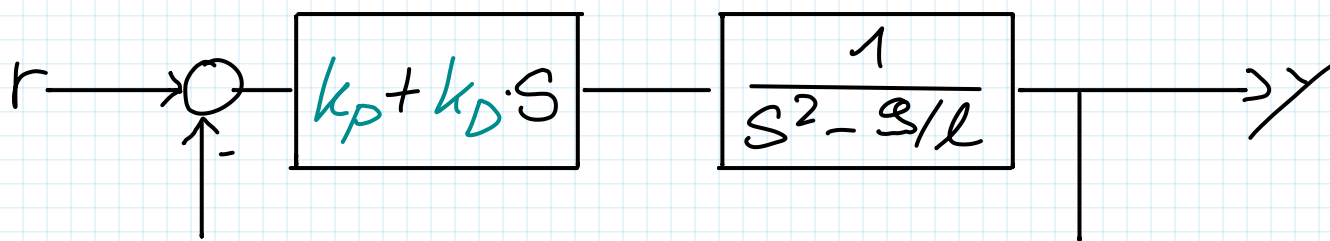
To find out if we can make this system asymptotically stable, we draw the rl:



Poles will stay on Im-axis.

$\Rightarrow$ Never asymptotically stable...

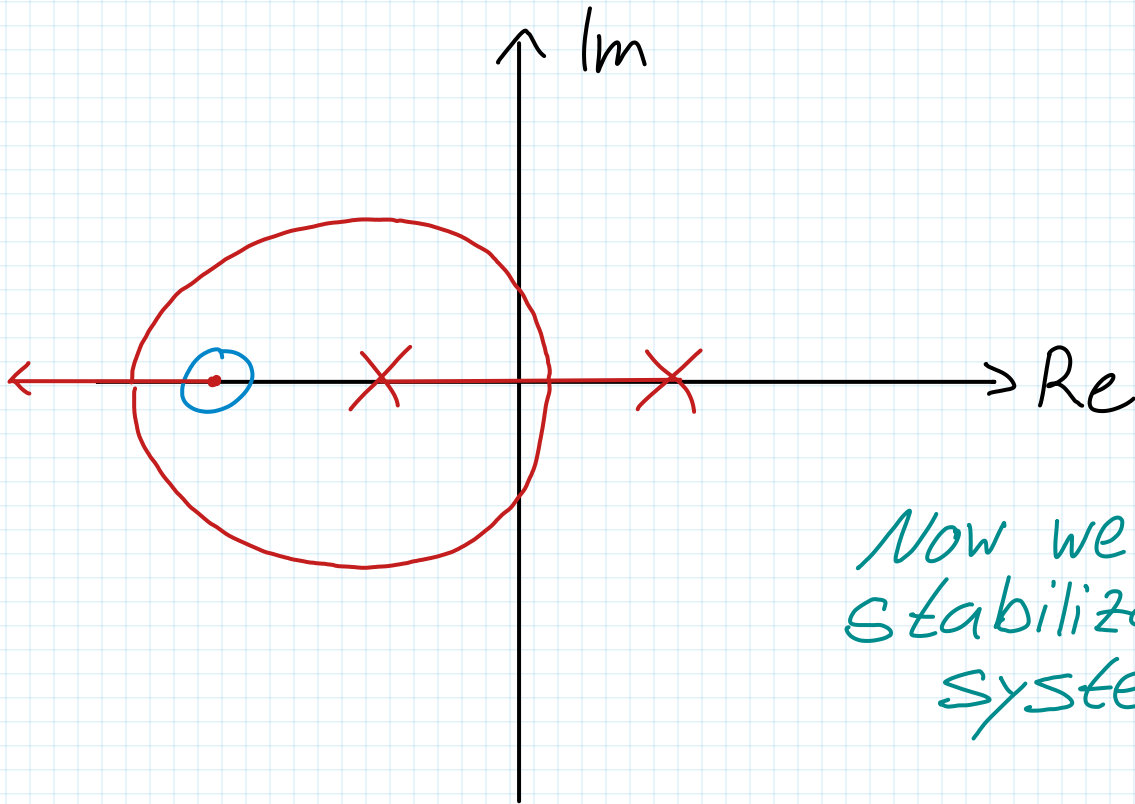
What if we choose a PD-controller?



We bring $L(s)$ into rl-form:

$$L(s) = \frac{k_p + k_D s}{s^2 - g/l} = \underbrace{k_D}_{K_{rl}} \cdot \frac{s + k_p/k_D}{(s - \sqrt{g/l})(s + \sqrt{g/l})}$$

Drawing root-locus again:

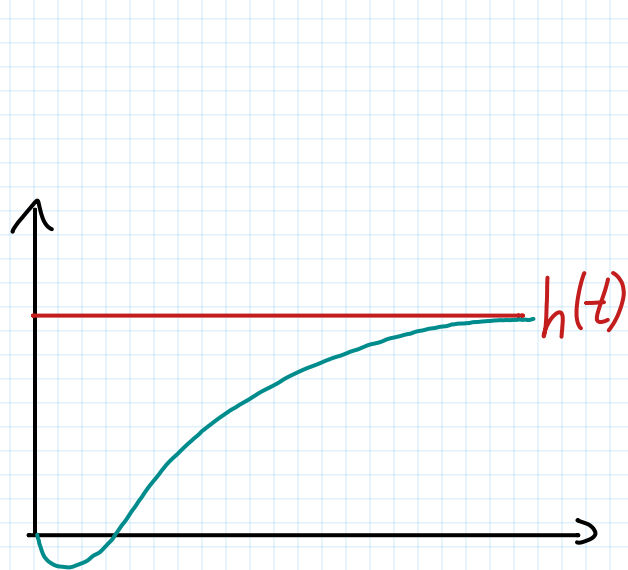
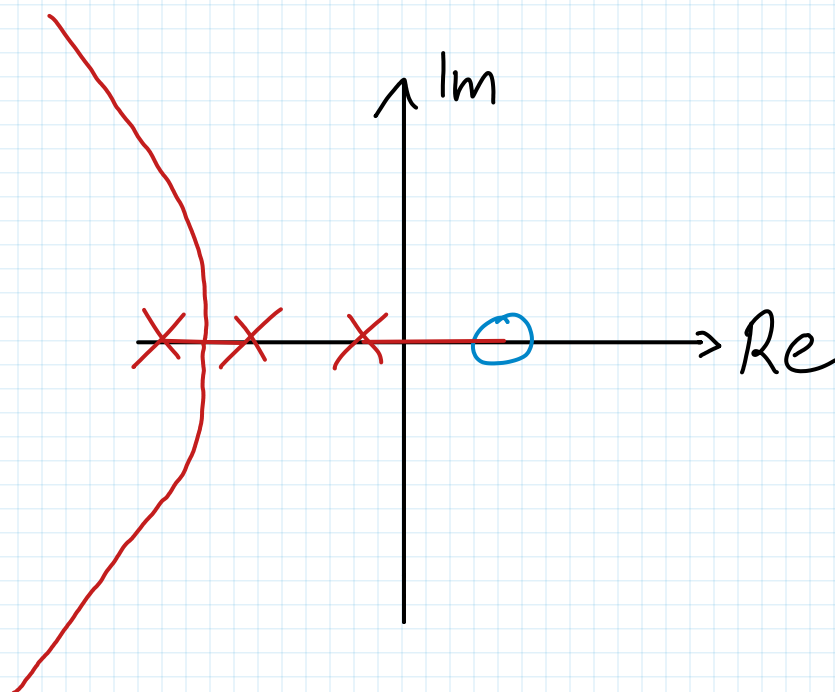


Now we can
stabilize the
system!!

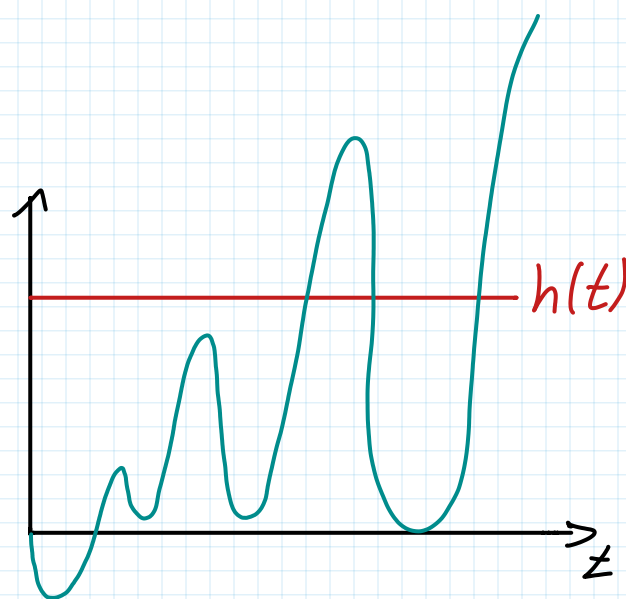
Almost done!

=> Quiz question!

Q3) Given the open-loop pole-zero plot, find the closed loop step responses for a very small, and a very large proportional gain. (hard)



$k \hat{=} \text{small}$



$k \hat{=} \text{large}$

